Problem Set 3

Applied Stats/Quant Methods 1/Nour Kebbi/ Answer Sheet

Due: November 13, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in R using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

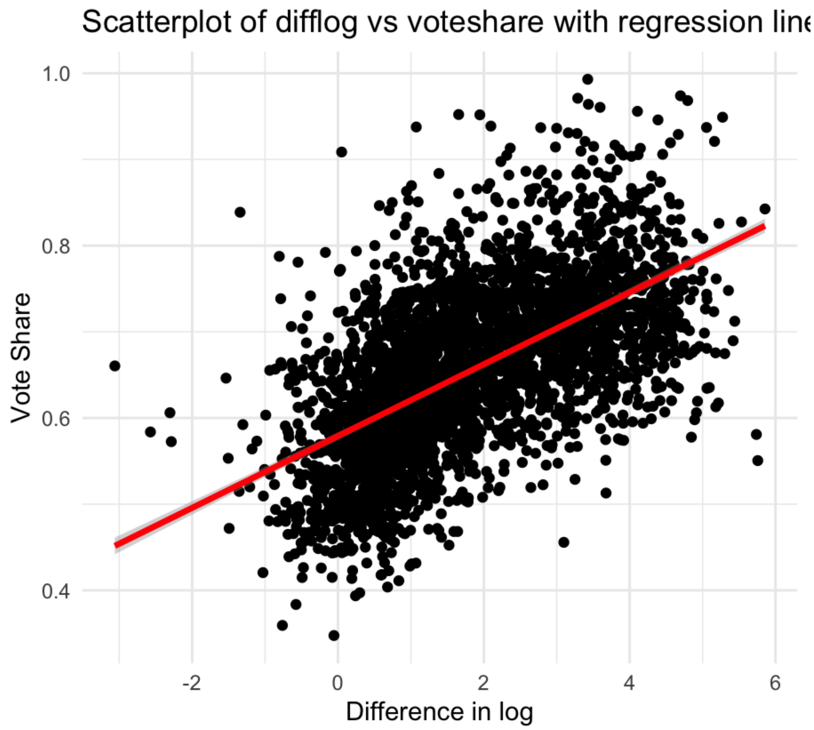
```
#runing the regression
model1 <- lm(voteshare ~ difflog, data = inc.sub)
#seeing the summary
summary(model1)
```

2. Make a scatterplot of the two variables and add the regression line.

```
#Part 2 scatterplot with regression line
scatter1 <- ggplot(inc.sub, aes(x = difflog, y = voteshare)) +
  geom_point() +                    # points
  geom_smooth(method = "lm", col = "red", lwd = 1.2) +
```

```
  labs(x = "Differencelog", y = "Voteshare",
       title = "Scatterplot of difflog vs voteshare with regression line"
  ) +
  theme_minimal()
```

Figure 1: Scatterplot of difflog vs voteshare with regression line:



3. Save the residuals of the model in a separate object.

```
#Part 3 — saving residuals in a seperate object
residuals_model1 <- residuals(model1)
#head(residuals_model1)
```

4. Write the prediction equation.

```
#Part 4 — prediction equation
coef_model1 <- coef(model1)
#coef_model1
intercept <- coef_model1[1]
slope <- coef_model1[2]
#writing prediction equation below
cat("Prediction equation: voteshare =", round(intercept, 3), "+", round(
    slope, 3), "* difflog\n")
```

Interpretation: on average, for every 1 unit increase in the log difference of spending, the incumbent's vote share is expected to increase by 0.024. The vote share without the difflog is 0.508

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

```r
#running the regression
model2 <- lm(presvote ~ difflog, data = inc.sub)
#seeing the summary
summary(model2)
```

2. Make a scatterplot of the two variables and add the regression line.

```r
#Part 2 scatterplot with regression line
scatter2 <- ggplot(inc.sub, aes(x = difflog, y = presvote)) +
  geom_point() +                          # points
  geom_smooth(method = "lm", col = "red", lwd = 1.2) +
  labs(x = "Differencelog", y = "presvote",
       title = "Scatterplot of difflog vs presvote with regression line")
    +
  theme_minimal()
```

3. Save the residuals of the model in a separate object.

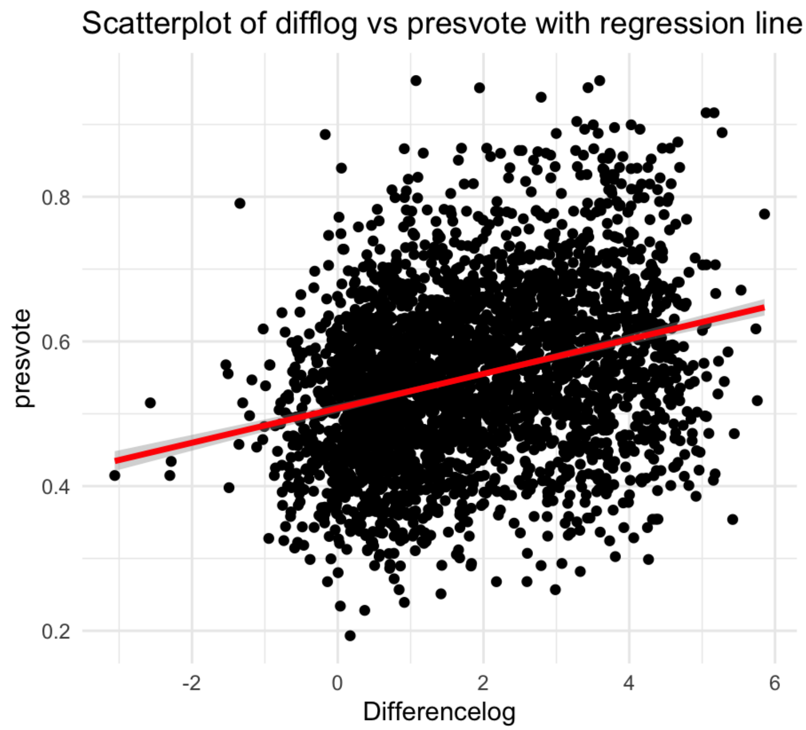
```r
#Part 3 - saving residuals in a seperate object
residuals_model2 <- residuals(model2)
#head(residuals_model1)
```

4. Write the prediction equation.

```r
#Part 4 - prediction equation
coef_model2 <- coef(model2)
#coef_model2
intercept <- coef_model2[1]
slope <- coef_model2[2]
#writing prediction equation below
cat("Prediction equation: presvote =", round(intercept, 3), "+", round(
    slope, 3), "* difflog\n")
```

Interpretation: on average, for every 1 unit increase in the log difference of spending, the pres vote is expected to increase by 0.024. The pres vote without the difflog is 0.508

Figure 2: Scatterplot of difflog vs presvote with regression line:



Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

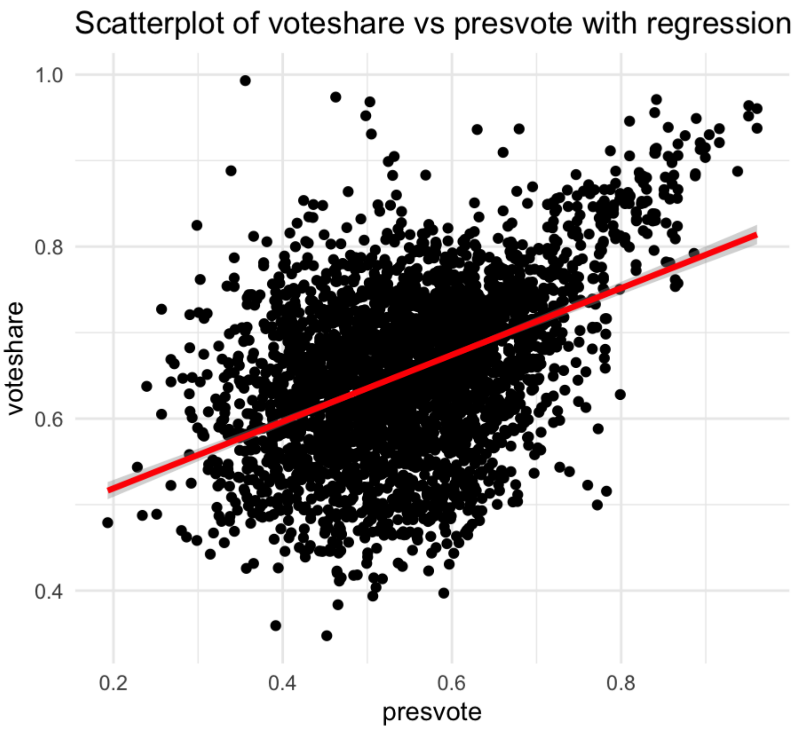
1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

```
#Part 1
#runing the regression
model3 <- lm(voteshare ~ presvote, data = inc.sub)
#seeing the summary
summary(model3)
```

2. Make a scatterplot of the two variables and add the regression line.

```
#Part 2 scatterplot with regression line
scatter3 <- ggplot(inc.sub, aes(x = presvote, y = voteshare)) +
  geom_point() +                          # points
  geom_smooth(method = "lm", col = "red", lwd = 1.2) +
  labs(x = "presvote", y = "voteshare",
       title = "Scatterplot of voteshare vs presvote with regression line
  ") +
  theme_minimal()
```

Figure 3: Scatterplot of voteshare vs presvote with regression line:



3. Write the prediction equation.

```r
#Part 3 - prediction equation
coef_model3 <- coef(model3)
#coef_model3
intercept <- coef_model3[1]
slope <- coef_model3[2]
#writing prediction equation below
cat("Prediction equation: voteshare =", round(intercept, 3), "+", round(
    slope, 3), "* presvote\n")
```

Interpretation: on average, for every 1 unit increase in the presvote, the vote share is expected to increase by 0.388 (slope). The vote share without the presvote is predicted to be 0.441 (intercept)

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
#runing the regression
model4 <- lm(residuals_model1 ~ residuals_model2)
#seeing the summary
summary(model4)
```

2. Make a scatterplot of the two residuals and add the regression line.

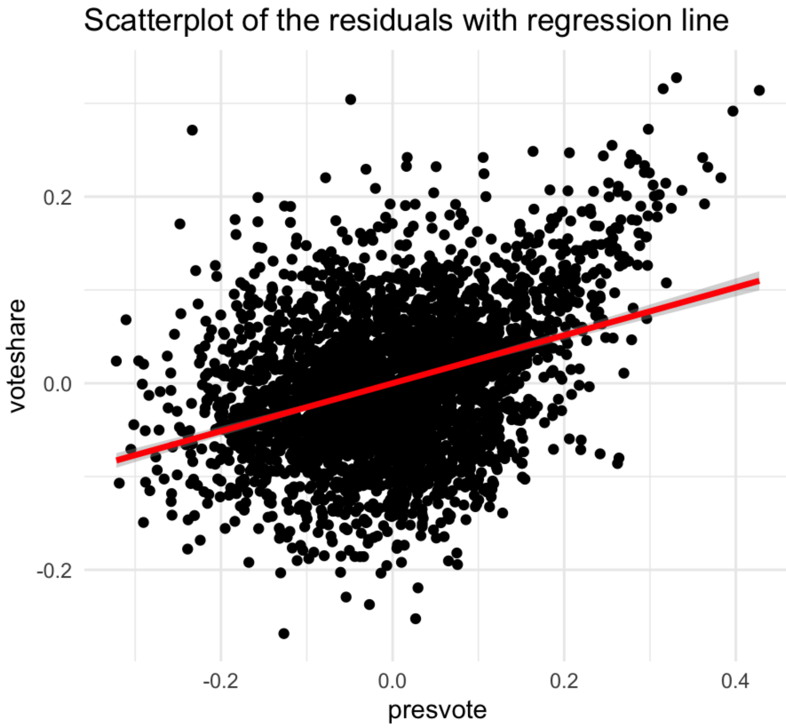
```
#Part 2 scatterplot with regression line
residuals_df <- data.frame(res1 = residuals_model1, res2 = residuals_
    model2)
scatter4 <- ggplot(residuals_df, aes(x = res2, y = res1)) +
  geom_point() +                              # points
  geom_smooth(method = "lm", col = "red", lwd = 1.2) +
  labs(x = "presvote", y = "voteshare",
       title = "Scatterplot of the residuals with regression line") +
  theme_minimal()
```

3. Write the prediction equation.

```
#Part 3 - prediction equation
coef_model4 <- coef(model4)
#coef_model4
intercept <- coef_model4[1]
slope <- coef_model4[2]
#writing prediction equation below
cat("Prediction equation: residuals in model 1 =", round(intercept, 3), "
   +", round(slope, 3), "* residuals in model 2\n")
```

Interpretation: on average, for every 1 unit increase in the residuals in model 2, the residuals in model 1 is expected to increase by 0.257 (slope). The residuals in model 1 without the ones in model 2 is predicted to be 0 (intercept)

Figure 4: Scatterplot of the residuals with regression line:



Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
#Part 1 - multiple regression
model5 <- lm(voteshare ~ difflog + presvote, data = inc.sub)
#seeing the summary
summary(model5)
```

2. Write the prediction equation.

```
#Part 2 - prediction equation
coef_model5 <- coef(model5)
#coef_model5
intercept <- coef_model5[1]
difflog <- coef_model5[2]
presvote <- coef_model5[3]
#writing prediction equation below
cat("Prediction equation: voteshare =", round(intercept, 3), "+", round(
    difflog, 3), "* difflog +", round(presvote, 3), "* presvote\n")
```

Interpretation: on average, for every 1 unit increase in the difflog, the voteshare is expected to increase by 0.0355 (slope) and on average, for every 1 unit increase in the presvote, the voteshare is expected to increase by 0.256 (slope2). The voteshare without both variable is predicted to be 0.448 (intercept)

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case? The slope is the same as in question 4 which is 0.449. This is because the slope in multiple regression of multiple variable is the same as the slope of the residuals of the effect of thee 2 variables on the outcome